

Multivariable Calculus

Labix

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1 Metric Properties of Euclidean Space

1.1 Component-wise Properties

Theorem 1.1.1 (Bolzano-Weierstrass Theorem) Any bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof Let $\{x_n\} \subset \mathbb{R}^n$ be bounded. By Bolzano-Weierstrass Theorem on $\mathbb{R}$, the sequence $\{x_{1,m}\} \subset \mathbb{R}$ has a convergent subsequence $\{x_{1,m_k}\}$. Apply Bolzano-Weierstrass Theorem to $\{x_{2,m_k}\}$ and keep going until you reach the n th component. We will end up with a subsequence that converges for all components and thus is convergent in $\mathbb{R}^n$. ■

Proposition 1.1.2

Let $m \in \mathbb{N}$. Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences in $\mathbb{R}^m$. Suppose that $(x_n)_{n \in \mathbb{N}}$ converges to $x \in \mathbb{R}^m$ and $(y_n)_{n \in \mathbb{N}}$ converges to $y \in \mathbb{R}^m$. Then the following are true.

- If $\lambda, \mu \in \mathbb{R}$, then $(\lambda x_n + \mu y_n)_{n \in \mathbb{N}}$ converges to $\lambda x + \mu y$.
- $(x_n y_n)_{n \in \mathbb{N}}$ converges to xy .

Theorem 1.1.3 (Componentwise Continuity) Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function. Then f is continuous if and only if each of its components $f_1, \dots, f_n : U \rightarrow \mathbb{R}$ is continuous.

Proof Suppose that f is continuous at $a \in \mathbb{R}^n$. Let $\epsilon > 0$. Then there exists $\delta > 0$ from continuity such that $\|x - a\| < \delta$ implies $\|f(x) - f(a)\| < \epsilon$. Then $\|x - a\| < \delta$ implies

$$\|f_i(x) - f_i(a)\| \leq \|f(x) - f(a)\| < \epsilon$$

Thus each component is continuous.

Now let f be component wise continuous at $a \in \mathbb{R}^n$. Let $\epsilon > 0$. For each component $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, using continuity with $\frac{\epsilon}{\sqrt{n}} > 0$ there exists $\delta_i > 0$ such that $\|x - a\| < \delta$ implies $\|f_i(x) - f_i(a)\| < \frac{\epsilon}{\sqrt{n}}$. Choose $\delta = \max\{\delta_1, \dots, \delta_n\}$. Then $\|x - a\| < \delta$ implies that

$$\begin{aligned} \|f(x) - f(a)\|^2 &= \sum_{k=1}^n \|f_k(x) - f_k(a)\|^2 \\ &\leq n \max_{k \in \{1, \dots, n\}} \|f_k(x) - f_k(a)\|^2 \\ \|f(x) - f(a)\| &\leq \sqrt{n} \max_{k \in \{1, \dots, n\}} \|f_k(x) - f_k(a)\| \\ &\leq \sqrt{n} \frac{\epsilon}{\sqrt{n}} \\ &= \epsilon \end{aligned}$$

Thus f is continuous. ■

Proposition 1.1.4 Let $U \subseteq \mathbb{R}^n$ be a subset. Let $f, g : U \rightarrow \mathbb{R}^m$ be continuous functions. Then the following are true.

- If $\lambda, \mu \in \mathbb{R}$, then $\lambda f + \mu g$ is continuous.
- fg is continuous.
- If $g(x) \neq 0$ for all $x \in U$, then $\frac{f}{g}$ is continuous.

Definition 1.1.5 (Limits)

Let $U \subseteq \mathbb{R}^n$ be a subset. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Let $c \in U$. We say that the limit of f at c is $L \in \mathbb{R}^m$ if for all $\epsilon > 0$, there exists $\delta > 0$ such that $|x - c| < \delta$ implies

$$|f(x) - L| < \epsilon$$

In this case, we write $L = \lim_{x \rightarrow c} f(x)$.

Lemma 1.1.6

Let $U \subseteq \mathbb{R}^n$ be a subset. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Let $c \in U$. Let $L = (L_1, \dots, L_m) \in \mathbb{R}^m$. Suppose that f is given by $f(x) = (f_1(x), \dots, f_m(x))$ for $f_i : U \rightarrow \mathbb{R}$. Then $\lim_{x \rightarrow c} f(x) = L$ if and only if $\lim_{x \rightarrow c} f_i(x) = L_i$ for $1 \leq i \leq m$.

Lemma 1.1.7

Let $U \subseteq \mathbb{R}^n$ be a subset. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Let $c \in U$. Then f is continuous at c if and only if $\lim_{x \rightarrow c} f(x) = f(c)$.

Theorem 1.1.8 (Extreme Value Theorem)

Let $n, m \in \mathbb{N} \setminus \{0\}$. Let $K \subseteq \mathbb{R}^n$ be a closed and bounded subset. Let $f : K \rightarrow \mathbb{R}^m$ be a function. Then there exists $a, b \in K$ such that

$$f(a) \leq f(x) \leq f(b)$$

for all $x \in K$.

Proof We know that $f(K)$ is closed and bounded. Thus it must have a supremum M and infimum m that are finite. Then we must have $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ such that $a_n \rightarrow m$ and $b_n \rightarrow M$ by definition of supremum and infimum. Since $f(K)$ is closed, $m, M \in f(K)$ and we are done. ■

1.2 Metric Structure on The Space of Linear Maps

In the discussion of derivatives of functions of the form $f : U \rightarrow \mathbb{R}^n \rightarrow \mathbb{R}^m$, we will need to understand maps of the form $U \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ and $U \rightarrow M_{m \times n}(\mathbb{R})$. Recall from Linear Algebra that the codomain of both types of maps are isomorphic to $\mathbb{R}^{n \times m}$, and through the isomorphism we may introduce the notion of metrics and hence continuity to these maps. In order to understand the metric structure more explicitly, we shall define the metric structures of each of them more explicitly, and discuss their relations.

Recall from linear algebra that given a choice of basis $e_1, \dots, e_n$ of $\mathbb{R}^n$ and a choice of basis $f_1, \dots, f_m$ of $\mathbb{R}^m$, we can construct an explicit vector space isomorphism

$$\text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m) \cong M_{m \times n}(\mathbb{R})$$

as follows. Given a linear map T , we may write $T(e_i) = \sum_{k=1}^m a_{ki} f_k$ for $a_{ki} \in \mathbb{R}$. Then the isomorphism is given by

$$T \mapsto \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

We use the $\mu : \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m) \xrightarrow{\cong} M_{m \times n}(\mathbb{R})$ to denote the isomorphism when $e_1, \dots, e_n$ and $f_1, \dots, f_m$ are the standard basis vectors for $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively.

Definition 1.2.1 (Operator Norm)

Define the operator norm of the vector space $\text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ to be the function $\|\cdot\| : \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m) \rightarrow \mathbb{R}$ given by

$$\|T\| = \sup\{|T(x)| \mid x \in \mathbb{R}^n \text{ such that } |x| = 1\}$$

Lemma 1.2.2 Let $T \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^k)$ and $S \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^k, \mathbb{R}^m)$. Then

$$\|S \circ T\| \leq \|S\| \|T\|$$

Proof We have that

$$\begin{aligned} |S(T(x))| &\leq \|S\| |T(x)| \\ &\leq \|S\| \|T\| |x| \end{aligned}$$

Thus we are done. ■

Definition 1.2.3 (The Operator Metric)

Define the operator metric $d : \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m) \times \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m) \rightarrow \mathbb{R}$ by the formula

$$d(T, S) = \|T - S\|$$

This gives a metric structure on $\text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ and we can now talk about functions $f : \mathbb{R}^k \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ which are continuous. Similarly, we would like to talk about continuous functions of the form $\mathbb{R}^k \rightarrow M_{m \times n}(\mathbb{R})$.

Definition 1.2.4 (The Frobenius Norm)

Define the Frobenius norm of the vector space $M_{m \times n}(\mathbb{R})$ to be the function $\|\cdot\| : M_{m \times n}(\mathbb{R}) \rightarrow \mathbb{R}$ given by

$$\|A\|_F = \left(\sum_{j=1}^n \sum_{i=1}^m a_{ij}^2 \right)^{\frac{1}{2}}$$

Definition 1.2.5 (The Frobenius Metric)

Define the operator metric $d : M_{m \times n}(\mathbb{R}) \times M_{m \times n}(\mathbb{R}) \rightarrow \mathbb{R}$ by the formula

$$d(A, B) = \|A - B\|_F$$

Lemma 1.2.6

Let $T \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^k)$ be a linear map. Let A be the matrix of T with respect to the standard basis. Then we have

$$\frac{1}{\sqrt{n}} \|A\|_F \leq \|T\| \leq \|A\|_F$$

Proof ■

Lemma 1.2.7

Let $U \subseteq \mathbb{R}^k$ be an open subset. Let $f : U \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ be a function. Then f is continuous if and only if $\mu \circ f : U \rightarrow M_{m \times n}(\mathbb{R})$ is continuous.

Proof ■

This lemma will give us our condition for continuous derivatives in terms of the Jacobian matrix later. However, we may also interpret this lemma as follows: To check the continuity of a map of the form $U \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$, it suffices to check that each term in the matrix function $U \rightarrow M_{m \times n}(\mathbb{R})$ is continuous. This is a practical way of checking that functions of the given type are continuous.

2 Differentiation

2.1 Frechet Derivative

Definition 2.1.1 (Frechet Derivatives)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $x \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. We say that f is differentiable at x if there exists a linear map $Df(x) \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ such that

$$\lim_{h \rightarrow 0} \frac{|f(x+h) - f(x) - Df(h)|}{|h|} = 0$$

In this case, we say that the derivative of f at x is $Df(x)$.

Proposition 2.1.2

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $x \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Suppose that f is differentiable at x with derivatives A and B in $\text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$. Then $A = B$.

Proof Suppose that A, B both represent the derivative of $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$. Then fix $\epsilon > 0$. There exists δ_1, δ_2 such that $|h| < \min\{\delta_1, \delta_2\}$ implies

$$\frac{|f(x+h) - f(x) - Ah|}{|h|} < \epsilon$$

and

$$\frac{|f(x+h) - f(x) - Bh|}{|h|} < \epsilon$$

Then we have

$$\begin{aligned} |(A - B)h| &= |f(x+h) - f(x) - Bh - f(x+h) - f(x) - Ah| \\ &\leq |f(x+h) - f(x) - Bh| + |f(x+h) - f(x) - Ah| \\ &\leq \epsilon|h| + \epsilon|h| \\ &= 2\epsilon|h| \end{aligned}$$

Thus $A = B$ by definition of limit and we are done. ■

Proposition 2.1.3

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $x \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. If f is differentiable at x , then f is continuous at x .

Proof Since f is continuous at x , there exists $A \in \mathbb{R}^{m \times n}$ such that

$$\lim_{h \rightarrow 0} \frac{|f(x+h) - f(x) - Ah|}{|h|} = 0$$

Fix $\epsilon > 0$, there exists $\delta_1 > 0$ such that $|h| < \delta_1$

$$\begin{aligned} |f(x+h) - f(x) - Ah| &\leq \epsilon|h| \\ |f(x+h) - f(x)| &\leq |Ah| + \epsilon|h| \\ &< (\|A\| + \epsilon)|h| \end{aligned}$$

Set $\delta_2 = \min \left\{ \delta_1, \frac{\epsilon}{\|A\| + \epsilon} \right\}$. Then $|h| < \delta_2$ implies

$$|f(x+h) - f(x)| < (\|A\| + \epsilon)\delta_2 < \epsilon$$

and we are done. ■

Similar to continuity, differentiability in higher dimensions can be broken down by its individual components. This way we only need to show individual differentiability to save trouble.

Proposition 2.1.4 (Componentwise Differentiability)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $x \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function whose components are given by $f = (f_1, \dots, f_m)$. Then f is differentiable at x if and only if $f_i : U \rightarrow \mathbb{R}$ is differentiable at x for $1 \leq i \leq m$.

Proposition 2.1.5

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $f : U \rightarrow \mathbb{R}^m$ be a constant function. Then the following are true.

- f is differentiable at U .
- For any $x \in U$, we have $Df(x) = 0$.

Proof Suppose that $f(x) = k$ for some $k \in \mathbb{R}^m$. Then I claim that $Df(x) = 0$. Indeed since

$$\lim_{h \rightarrow 0} \frac{|0 + 0 - 0h|}{|h|} = 0$$

thus the definition of differentiability is satisfied. ■

Proposition 2.1.6

Let $A \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ be a linear map. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the function defined by $f(x) = Ax$. Then the following are true.

- f is differentiable at $\mathbb{R}^n$.
- For any $x \in \mathbb{R}^n$, we have $Df(x) = A$.

Proof Clearly $Df(x) = A$ satisfies the definition of differentiability since

$$\lim_{h \rightarrow 0} \frac{|A(x+h) - Ax - Ah|}{|h|} = 0$$

Translation: Linear maps have constant derivatives (as in the one dimensional case).

2.2 The Jacobian Matrix Representing the Derivative

Definition 2.2.1 (Partial Derivatives)

Let $e_1, \dots, e_n$ be the standard basis for $\mathbb{R}^n$. Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $p \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function whose components are given by $f = (f_1, \dots, f_m)$. Define the j th partial derivative of the i th component of f at p to be

$$\left. \frac{\partial f_i}{\partial x_j} \right|_p = \lim_{t \rightarrow 0} \frac{f_i(p + te_j) - f_i(p)}{t} = \left. \frac{d}{dt} f(p + te_j) \right|_{t=0}$$

Note that for a function $f : U \rightarrow \mathbb{R}^m$, the partial derivative $\left. \frac{\partial f_i}{\partial x_j} \right|_p$ is a real number.

Example 2.2.2

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) = \begin{cases} \frac{xy}{(x^2 + y^2)^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } x = y = 0 \end{cases}$$

Then the following are true.

- The 1st partial derivative of f at (x, y) is given by

$$\partial_{(1,0)} f(x, y) = \frac{y}{(x^2 + y^2)} - \frac{4x^2 y}{(x^2 + y^2)^3}$$

for $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

- The 2nd partial derivative of f at (x, y) is given by

?

for $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

Definition 2.2.3 (Jacobian Matrix)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $p \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Define the Jacobian of f at p to be the matrix

$$Jf(p) = \begin{pmatrix} \left. \frac{\partial f_1}{\partial x_1} \right|_p & \cdots & \left. \frac{\partial f_1}{\partial x_n} \right|_p \\ \vdots & \ddots & \vdots \\ \left. \frac{\partial f_m}{\partial x_1} \right|_p & \cdots & \left. \frac{\partial f_m}{\partial x_n} \right|_p \end{pmatrix}$$

Proposition 2.2.4

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $p \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. If f is differentiable at p , then we have

$$Df(x)v = Jf(x)v$$

for all $v \in \mathbb{R}^n$.

Proof Note that we have

$$\begin{aligned} Df(x)v &= Df(x) \sum_{k=1}^n v_k e_k \\ &= \sum_{k=1}^n v_k Df(x)e_k \\ &= \sum_{k=1}^n v_k \partial_k f(x) \\ &= \partial f(x)v \end{aligned}$$

■

In other words, the Jacobian matrix $Jf(x) \in M_{m \times n}(\mathbb{R})$ is the representation of the linear map $Df(x) \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ under the standard basis for $\mathbb{R}^n$ and $\mathbb{R}^m$.

Proposition 2.2.5

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $B_r(x_0) \subseteq U$ be a subset for some $r > 0$ and $x_0 \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Suppose that the following are true.

- The Jacobian $Jf(x)$ exists for all $x \in B_r(x_0)$.
- $Jf : B_r(x_0) \rightarrow M_{m \times n}(\mathbb{R})$ is continuous.

Then f is differentiable.

Proof We show this for the case that $k = 1$ since $f : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is differentiable if and only if each component $f_1, \dots, f_k$ is differentiable. Thus now $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Suppose that ∂f is continuous in $B_r(x)$ for fixed x . This means that $\partial_{x_j} f_i$ is continuous for any $j \in \{1, \dots, n\}$ and $i \in \{1, \dots, k\}$.

Take a point $x + h \in B_r(x)$. Define $h = \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}$ and

$$h'_k = \begin{pmatrix} h_1 \\ \vdots \\ h_k \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Trivially define $h'_0 = 0$ and $h_0 = 0$ for convenience. Now $f(x + h'_k) - f(x + h'_{k-1})$ is just a one variable function on the k th slot, thus we can apply the mean value theorem in real analysis and conclude that there exists $\theta_k \in (0, 1)$ such that $f(x + h'_k) - f(x + h'_{k-1}) = \partial_{x_k} f(x + h'_{k-1} + (\theta_k h_k) e_k) h_k$.

Summing all these up, we have that

$$\begin{aligned} \sum_{k=1}^n f(x + h'_k) - f(x + h'_{k-1}) &= \sum_{k=1}^n \partial_{x_k} f(x + h'_{k-1} + (\theta_k h_k) e_k) h_k \\ f(x + h) - f(x) &= \sum_{k=1}^n \partial_{x_k} f(x + h'_{k-1} + (\theta_k h_k) e_k) h_k \end{aligned}$$

We now use the continuity of the partial derivatives. Given $\epsilon > 0$, there $\delta_k > 0$ such that $|h| < \delta_k$ implies $|\partial_{x_k} f(x + h) - \partial_{x_k} f(x)| < \epsilon$. Choose $\delta = \min\{\delta_1, \dots, \delta_n\}$. Since $\theta_k \in (0, 1)$ for all $k \in \{1, \dots, n\}$, we must have $|h'_{k-1} + (\theta_k h_k) e_k| < |h| < \delta$. Thus

$$\begin{aligned} \left| f(x + h) - f(x) - \sum_{k=1}^n \partial_{x_k} f(x) h_k \right| &= \left| \sum_{k=1}^n \partial_{x_k} f(x + h'_{k-1} + (\theta_k h_k) e_k) h_k - \sum_{k=1}^n \partial_{x_k} f(x) h_k \right| \\ &< \epsilon \sum_{k=1}^n |h_k| \\ &< \epsilon \sqrt{n} |h| \end{aligned}$$

if $|h| < \delta$. The second inequality is due to the fact that $\sum_{k=1}^n h_k \leq \sqrt{n} |h|$.

Now define $A \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$ by $A(h) = (\partial f)h$. This makes sense in matrix multiplication since $\partial f \in M_{1 \times n}(\mathbb{R})$. Then $0 < |h| < \delta$ implies

$$\begin{aligned} \frac{|f(x + h) - f(x) - Ah|}{|h|} &\leq \epsilon \sqrt{n} \frac{|h|}{|h|} \\ &= \epsilon \sqrt{n} \end{aligned}$$

since $A(h) = \sum_{k=1}^n \partial_{x_k} f(x) h_k$. Since this is true for all ϵ , the condition for differentiability is satisfied and we are done. ■

Notice that in the above proof, in order to show that f is differentiable at x , we need to know the partial derivatives of its neighbours.

The below theorem acts as a summary or recollection of the relationships between differentiability and continuous Jacobian Matrices.

Corollary 2.2.6

Let U be an open subset of $\mathbb{R}^n$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Then f is continuously differentiable if and only if $Jf : U \rightarrow M_{m \times n}(\mathbb{R})$ is continuous.

2.3 Directional Derivatives**Definition 2.3.1** (Directional Derivative)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $p \in U$ be a point. Let $f : U \rightarrow \mathbb{R}^m$ be a function. Let $v \in \mathbb{R}^n$ be a direction. Define the directional derivative of f in the direction v at p to be

$$\partial_v f(p) = \lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t} = \left. \frac{d}{dt} f(p + tv) \right|_{t=0}$$

if the limit exists.

Note that $\partial_v f(p)$ is a vector in $\mathbb{R}^m$.

Lemma 2.3.2

Let $e_1, \dots, e_n$ be the standard basis for $\mathbb{R}^n$. Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $p \in U$ be a point. Let $f : U \rightarrow \mathbb{R}^m$ be a function whose components are given by $f = (f_1, \dots, f_m)$. Then we have

$$\partial_{e_j} f(p) = \left(\left. \frac{\partial f_1}{\partial x_j} \right|_p, \dots, \left. \frac{\partial f_m}{\partial x_j} \right|_p \right)$$

for $1 \leq j \leq n$.

Proposition 2.3.3

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $p \in U$. Let $f : U \rightarrow \mathbb{R}^m$ be a function. If f is differentiable at p , then the following are true.

- For any $v \in \mathbb{R}^n$, we have $Df(p)v = Jf(x)v = \partial_v f(x)$.
- For $\lambda, \mu \in \mathbb{R}$ and $v, w \in \mathbb{R}^n$, we have

$$\partial_{\lambda v + \mu w} f(x) = \lambda \partial_v f(x) + \mu \partial_w f(x)$$

$(\partial_v f(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m)$ is a linear map).

Proof Suppose that Df exists. Then using the definition of differentiability, we have that

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x) - Df(x)(tv)}{t|v|} &= 0 \\ \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x) - tDf(x)v}{t} &= 0 \\ \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t} &= \lim_{t \rightarrow 0} \frac{tDf(x)v}{t} \\ \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t} &= Df(x)v \\ \partial_v f(x) &= Df(x)v \end{aligned}$$

Linearity follows since $Df(x)$ is matrix and matrix multiplication on a vector is linear. ■

We should make absolutely clear distinction with the existence of frechet derivatives and the existence of directional derivatives. Even if the directional derivatives exists, as long as they are not linear, the matrix for the frechet derivative will not exist. In particular, it is possible to write out the Jacobian matrix as we see below, but this Jacobian matrix will not be equal to the matrix in the frechet derivative.

The above theorem is also useful for detecting non-differentiability. By verifying the directional derivatives are non-linear, one can use the contrapositive to prove that f would not be differentiable.

2.4 Properties of the Derivative

Theorem 2.4.1 (Algebra of Differentiable functions) If $f, g : U \rightarrow \mathbb{R}^m$ are differentiable functions at x and $\lambda, \mu \in \mathbb{R}$ are constants, then

$$D(\lambda f + \mu g)(x) = \lambda Df(x) + \mu Dg(x)$$

Lemma 2.4.2 Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^k$, $x \in U$ and $r > 0$ such that $B_r(x) \subset U$ and $A \in L(\mathbb{R}^n, \mathbb{R}^k)$. Define $\Delta_{x,A}f : B_r(0) \rightarrow \mathbb{R}^k$ by

$$\Delta_{x,A}f(h) = \begin{cases} \frac{f(x+h)-f(x)-Ah}{|h|} & \text{if } h \neq 0 \\ 0 & \text{if } h = 0 \end{cases}$$

Then f is differentiable at x with $Df(x) = A$ if and only if $\Delta_{x,A}$ is continuous at 0.

Proof Let $\Delta_{x,A}$ be continuous at 0. Then

$$\lim_{h \rightarrow 0} |\Delta_{x,A}f(h)| = |\Delta_{x,A}f(0)| = 0$$

Thus by definition of differentiability f is differentiable at x with $Df(x) = A$.

Now let f be differentiable at x and set $A = Df(x)$. Then by definition of differentiability we have that

$$\lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x) - Ah}{|h|} \right| = \lim_{h \rightarrow 0} |\Delta_{x,A}f(h)| = 0 = \Delta_{x,A}f(0)$$

Thus $\Delta_{x,A}f(x)$ is continuous at 0. ■

Lemma 2.4.3 Let $\tau > 0$. Let $\xi : B_\tau(0) \rightarrow \mathbb{R}$ be bounded and $\nu : B_\tau(0) \rightarrow \mathbb{R}^k$ be continuous at $0 \in B_\tau(0)$ and $\nu(0) = 0$. Then

$$\delta(h) = \xi(h)\nu(h)$$

where $0 < |h| < \tau$ and $\delta(0) = 0$ is continuous at $0 \in B_\tau(0)$.

Proof By continuity of ν at 0, let $\epsilon > 0$. Then there exists $\delta \in (0, \tau)$ such that $|h| < \tau$ implies $|\nu(h)| < \epsilon$. By boundedness of ξ , there exists $M > 0$ such that $|\xi| < M$ for all $h \in B_\tau(0) \setminus \{0\}$. Thus $0 < |h| < \delta$ implies $|\delta(h)| < M\epsilon$, meaning $\lim_{h \rightarrow 0} \delta(h) = 0 = \delta(0)$ thus we are done. ■

Proposition 2.4.4 (Chain Rule) Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : V \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$ be two differentiable functions. Let $x \in U$. Then $g \circ f$ is differentiable and

$$D(g \circ f)(x) = (Dg)(f(x)) \cdot Df(x)$$

Proof Let

$$\Delta_x f(h) = \begin{cases} \frac{f(x+h)-f(x)-Df(x)h}{|h|} & \text{if } h \neq 0 \\ 0 & \text{if } h = 0 \end{cases}$$

and

$$\Delta_{f(x)} g(k) = \begin{cases} \frac{g(f(x)+k)-g(f(x))-Dg(f(x))k}{|k|} & \text{if } k \neq 0 \\ 0 & \text{if } k = 0 \end{cases}$$

where both functions are continuous by lemma 2.3.2. Then we have that

$$f(x+h) = f(x) + Df(x)h + \Delta_x f(h)|h|$$

and

$$g(f(x)+k) = g(f(x)) + Dg(f(x))k + \Delta_{f(x)} g(k)|k|$$

Let $k(h) = Df(x)h + \Delta_x f(h)|h|$. Then by linearity of $Dg(f(x))$,

$$g(f(x+h)) = g(f(x)) + Dg(f(x))Df(x)h + Dg(f(x))\Delta_x f(h)|h| + \Delta_{f(x)} g(k(h))|k(h)|$$

Let

$$\delta_1(h) = D_g(f(x))(\Delta_x f(h))$$

and

$$\delta_2(h) = \begin{cases} \frac{|k(h)|}{|h|} \Delta_{f(x)} g(k(h)) & \text{if } h \neq 0 \\ 0 & \text{if } h = 0 \end{cases}$$

Then we have

$$g(f(x+h)) - g(f(x)) - D_g(f(x)) \circ Df(x)h = |h|(\delta_1(h) + \delta_2(h))$$

We now show that $\lim_{h \rightarrow 0} |\delta_1(h)| = 0$ and $\lim_{h \rightarrow 0} |\delta_2(h)| = 0$. Note that

$$|\delta_1(h)| \leq \|D_g(f(x))\| |\Delta_x f(h)|$$

Since $\lim_{h \rightarrow 0} |\Delta_x f(h)| = 0$ by construction, we are done. Now let

$$\xi(h) = \frac{|k(h)|}{|h|} \leq \frac{|Df(x)h|}{|h|} + |\Delta_x f(h)| \leq \|Df(x)\| + |\Delta_x f(h)|$$

Since $\Delta_x f$ is continuous, ξ is bounded on $B_r(0) \setminus \{0\}$ for some $r > 0$. Now set $\nu(h) = \Delta_{f(x)} g(k(h))$. Since k is continuous at $k(0) = 0$ and g is differentiable at $f(x)$, we have that $\nu(h)$ is continuous and $\nu(0) = 0$. Thus we can apply lemma 2.3.3 and

$$\lim_{h \rightarrow 0} |\delta_2(h)| = 0$$

Thus we must have

$$g(f(x+h)) - g(f(x)) - D_g f(x) \circ Df(x)h = |h|(\delta_1(h) + \delta_2(h)) \rightarrow 0$$

and we are done. ■

Corollary 2.4.5

Let $U, V \subseteq \mathbb{R}^n$ be open subsets. Let $p \in U$. Let $f : U \rightarrow V$ be a bijective function. Suppose that f is differentiable at p and f^{-1} is differentiable at $f(p)$. Then we have

$$Df^{-1}(p) = (Df(p))^{-1}$$

2.5 Mean Value Theorems

Theorem 2.5.1 (Mean Value Theorem) Let $x, y \in \mathbb{R}^n$. Let $r : [a, b] \rightarrow \mathbb{R}^n$ be continuously differentiable. Let $f : C^1(r([a, b]), \mathbb{R}^k)$ and there exists some M such that $|\partial f(x)| \leq M$ for all $x \in U$. Then

$$|f(r(b)) - f(r(a))| \leq M \int_a^b |r'(t)| dt$$

Proof We have that

$$\begin{aligned}
 |f(r(b)) - f(r(a))| &= \left| \int_a^b \frac{d}{dt} f(r(t)) dt \right| && \text{(By the FTC)} \\
 &= \left| \int_a^b \partial f(r(t)) r'(t) dt \right| \\
 &\leq \int_a^b |\partial f(r(t)) r'(t)| dt \\
 &\leq \int_a^b |\partial f(r(t))| |r'(t)| dt \\
 &\leq \int_a^b M |r'(t)| dt
 \end{aligned}$$

Thus we are done. ■

Notice that $\int_a^b |r'(t)| dt$ is exactly the length of the path r from a to b .

Proposition 2.5.2 Suppose that $U \subseteq \mathbb{R}^n$ is path-connected and every path is differentiable. Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable and that $\partial f(x) = 0$ for all $x \in U$. Then f is constant on U .

Proof Let $x, y \in U$, then any path $r : [a, b] \rightarrow U$ with $r(a) = x$ and $r(b) = y$ is differentiable. Then

$$\begin{aligned}
 f(y) - f(x) &= f(r(b)) - f(r(a)) \\
 &= \int_a^b \frac{d}{dt} f(r(t)) dt && \text{(By the FTC)} \\
 &= \int_a^b \partial f(r(t)) r'(t) dt \\
 &= 0
 \end{aligned}$$

Thus $f(x) = f(y)$ for any $x, y \in U$. ■

3 More Properties of Differentiability

3.1 Inverse Function Theorem

The inverse function theorem is a powerful multidimensional analog of finding the derivative of an inverse function. The function has a few conditions to satisfy in order for it to be applied. Since the proof is exceptionally long, I will split it to a number of theorems and prepositions.

Proposition 3.1.1 Suppose that $\Psi : U \rightarrow V$ is a bijection which is differentiable at $x \in U$. Suppose that Ψ^{-1} is differentiable at $y = \Psi(x) \in V$. Then $D\Psi(x)$ and $D\Psi^{-1}(y)$ are both invertible and

$$(D\Psi^{-1})(y) = (D\Psi(\Psi^{-1}(y)))^{-1}$$

Proof Differentiating the relation $\Psi(\Psi^{-1}(y)) = y$ gives

$$D\Psi(\Psi^{-1}(y)) \circ D\Psi^{-1}(y) = I_n$$

which is the identity transformation thus we are done. ■

Lemma 3.1.2 $T \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^k)$ is injective if and only if there exists $a > 0$ such that $|T(x)| \geq a|x|$ for all $x \in \mathbb{R}^n$.

Proof Firstly suppose that there exists $a > 0$ such that $|T(x)| \geq a|x|$ for all $x \in \mathbb{R}^n$. Let $T(x) = T(y)$. Then

$$\begin{aligned} |T(x - y)| &\geq a|x - y| \\ |T(x) - T(y)| &\geq a|x - y| \\ 0 &\geq a|x - y| \end{aligned}$$

Since $a > 0$ we must have $x = y$.

Now suppose that for all $a > 0$, $|T(x)| < a|x|$ for all $x \in \mathbb{R}^n$. This means that $\frac{|T(x)|}{|x|} < a$ for all $a > 0$. This is precisely the definition of a limit. We can find a sequence $(x_j)_{j \in \mathbb{N}}$ such that

$$\frac{|T(x_j)|}{|x_j|} \rightarrow 0$$

Now consider a new sequence defined by $y_j = \frac{x_j}{|x_j|}$. Clearly $(y_j)_{j \in \mathbb{N}} \subset S^{n-1}$. But S^{n-1} is compact by the Heine-Borel theorem. Thus there exists $(y_{j_m})_{m \in \mathbb{N}}$ such that it has its limit in S^{n-1} . But then

$$T(y_j) = \frac{T(x_j)}{|x_j|}$$

thus $|T(y_j)| \rightarrow 0$. All the y_j are clearly nonzero, this means that T has a non-trivial kernel and thus T is not injective. ■

We split the inverse function theorem into its injective and surjective part for better readability.

Proposition 3.1.3 (Injective Part of Inverse Function Theorem) Let U be an open subset of $\mathbb{R}^n$ and suppose that $\Psi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that it is differentiable and its derivative is continuous in U . Assume that $D\Psi(p)$ is injective at a point $p \in U$. Then there exists $\delta > 0$ such that $B_\delta(p) \subset U$ and such that f is injective on $B_\delta(p)$.

Proof From the above lemma, $D\Psi(p)$ being injective means that there exists $a > 0$ such that

$$|Df(p)h| \geq \epsilon|h|$$

for all $h \in \mathbb{R}^n$. Since $Df : U \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^k)$ is continuous, there exists $\delta > 0$ such that $x \in$

$B_\delta(p) \subset U$ implies

$$\|Df(p) - Df(x)\| < \frac{1}{2}\epsilon$$

Define a new function $F : U \rightarrow \mathbb{R}^k$ by $F(x) = f(x) - Df(p)x$. Then F is differentiable since $Df(p)x$ is just a linear transformation and we have

$$DF(x) = Df(x) - Df(p)$$

Thus we now have

$$\|DF(x)\| = \|Df(x) - Df(p)\| < \frac{1}{2}\epsilon$$

We can now apply the mean value inequality to get

$$|F(z) - F(x)| \leq \frac{1}{2}\epsilon|z - x|$$

for all $x, z \in B_\delta(p)$. Finally we have

$$\begin{aligned} |f(x) - f(z)| &= |Df(p)(x - z) - (F(z) - F(x))| \\ &\geq \epsilon|x - z| - \frac{1}{2}\epsilon|x - z| \\ &= \frac{1}{2}\epsilon|x - z| \end{aligned}$$

This means that $x \neq z$ implies $f(x) \neq f(z)$ and we are done. ■

Proposition 3.1.4 (Surjective Part of Inverse Function Theorem) Let U be an open subset of $\mathbb{R}^n$ and suppose that $\Psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that it is differentiable and its derivative is continuous in U . Assume that $D\Psi(p)$ is surjective at a point $p \in U$. Then there exists $\rho > 0$ such that $B_\rho(\Psi(p)) \subset \Psi(U)$.

Proof By the rank nullity theorem, $D\Psi(p)$ is injective (surjectivity implies bijectivity in linear maps). By the above proposition, there exists $\epsilon > 0$ such that

$$|D\Psi(p)h| \geq \epsilon|h|$$

for all $h \in \mathbb{R}^n$. Again define $F : U \rightarrow \mathbb{R}^n$ by $F(x) = \Psi(x) - D\Psi(p)x$. Then exactly the same as the above proof, we must have $|F(x) - F(z)| \leq \frac{1}{2}\epsilon|x - z|$ and $|\Psi(x) - \Psi(z)| \geq \frac{1}{2}\epsilon|x - z|$. Set

$$K = \overline{B_{\frac{1}{2}\delta}(p)} = \{x \in \mathbb{R}^n \mid |x - p| \leq \frac{1}{2}\delta\}$$

and $\partial K = \{x \in \mathbb{R}^n \mid |x - p| = \frac{1}{2}\delta\}$. Now we have

$$\begin{aligned} |\Psi(x) - \Psi(p)| &\geq \frac{1}{2}\epsilon|x - p| \\ &= \frac{1}{4}\epsilon\delta \end{aligned}$$

For all $x \in \partial K$. Set $\rho = \frac{1}{8}\epsilon\delta$ and fix $y \in B_\rho(\Psi(p))$. We will show that

$$y \in \Psi(B_{\frac{1}{2}\delta}(p))$$

to finish the proof. Define a new function $\phi : K \rightarrow \mathbb{R}$ by $\phi(x) = |\Psi(x) - y|$. Then ϕ is continuous by composition of continuous functions. By the extreme value theorem, there exists $c \in K$ such that $\phi(c) \leq \phi(x)$ for all $x \in K$.

Now I will show that $c \in B_{\frac{1}{2}\delta}(p)$ by showing that $c \notin \partial K$ and using the fact that $B_{\frac{1}{2}\delta}(p) = K \setminus \partial K$.

$c \notin \partial K$ can be proved by $\phi(x) > \phi(p)$ for all $x \in \partial K$. Now if $x \in \partial K$, we have

$$\begin{aligned}\phi(x) &= |\Psi(x) - y| \\ &\geq |\Psi(x) - \Psi(p)| - |y - \Psi(p)| \\ &\geq \frac{1}{4}\epsilon\delta - \frac{1}{8}\epsilon\delta \\ &= \rho \\ &> |y - \Psi(p)| \\ &= \phi(p)\end{aligned}$$

Thus we are done with this part.

Now since $D\Psi(p)$ is surjective, there exists $h \in \mathbb{R}^n$ such that $D\Psi(p)h = y - \Psi(c)$ (by nuisances). Since $c \in B_{\frac{1}{2}\delta}(p)$, there exists $\nu > 0$ such that $|t| < \nu$ implies $|c + th - p| < \frac{1}{2}\delta$. Thus we have

$$\begin{aligned}\Psi(c + th) - y &= \Psi(c + th) - \Psi(c) - tD\Psi(p)h + (\Psi(c) - y + tD\Psi(p)h) \\ &= F(c + th) - F(c) + (1 - t)(\Psi(c) - y)\end{aligned}$$

for $|t| < \nu$. Now, we have that

$$\begin{aligned}\phi(c) &\leq \phi(c + th) \\ &= |\Psi(c + th) - y| \\ &\leq \frac{1}{2}t|D\Psi(p)h| + (1 - t)|\Psi(c) - y| \\ &= \left(1 - \frac{1}{2}t\right)(\Psi(c) - y)\end{aligned}$$

Since this holds for all $|t| < \nu$, we must have that $\phi(c) = 0$, which implies that $y = \Psi(c)$. Thus we have shown that

$$B_\rho(\Psi(p)) \subset \Psi(B_{\frac{1}{2}\delta}(p)) \subset \Psi(U)$$

■

Corollary 3.1.5 Let $U \subset \mathbb{R}^n$ be open. Let $\Psi \in \mathcal{C}^1(U, \mathbb{R}^n)$ and suppose that $D\Psi(p)$ is invertible for all points $p \in U$, then Ψ maps open subsets of U to open subsets of $\mathbb{R}^n$.

Proof Let $V \subset U$ be open. By the above proposition applied to $\Psi|_V$, we have for all $p \in V$, there exists $\rho > 0$ such that

$$B_\rho(\Psi(p)) \subset \Psi(V)$$

, thus $\Psi(V)$ is open. ■

Theorem 3.1.6 (Inverse Function Theorem) Let U be an open subset of $\mathbb{R}^n$ and suppose that $\Psi \in \mathcal{C}^1(U, \mathbb{R}^n)$. Assume that $D\Psi(p)$ is invertible at a point $p \in U$. Let $q = \Psi(p)$. Then there exists a neighbourhood N_p and N_q such that $\Psi : N_p \rightarrow N_q$ is a bijection and $\Psi^{-1} : N_q \rightarrow N_p$ is continuously differentiable and

$$(D\Psi^{-1})(y) = (D\Psi(\Psi^{-1}(y)))^{-1}$$

for all $y \in N_q$.

Proof Clearly Ψ satisfies the above two propositions. Thus this means that there exists $\epsilon > 0$ such that $|Df(p)h| \geq \epsilon|h|$ for all $h \in \mathbb{R}^n$ and that there exists $\delta > 0$ such that $x \in B_\delta(p) \subset U$ implies

$$\|Df(p) - Df(x)\| < \frac{1}{2}\epsilon$$

Using these two facts and the rank nullity theorem we deduce that $D\Psi(x)$ is invertible for all $x \in B_\delta(p)$. The above corollary implies that $\Psi(B_\delta(p))$ is open. Thus $\Psi|_{B_\delta(p)} : B_\delta(p) \rightarrow \Psi(B_\delta(p))$

is a bijection.

Now we show that Ψ^{-1} is continuously differentiable. ■

Note that there could be multiple p such that Ψ maps p to q . The inverse function theorem guarantees that as long as $D\Psi(p)$ is invertible, then they will have bijective neighbourhoods. This is why we can have multiple branches for the inverse. A good example would be $y = x^2$. It has two inverses, namely $\sqrt{x}$ and $-\sqrt{x}$ precisely because of this reason. In this case the two neighbourhoods would be $\mathbb{R}^+$ and $\mathbb{R}^-$ respectively.

Furthermore, even if there are multiple p mapping to q , not every p could have a neighbourhood such that they are bijective because we must require the fact that $D\Psi(p) \neq 0$.

3.2 Implicit Function Theorem

Theorem 3.2.1 (Implicit Function Theorem)

Let U be an open subset of $\mathbb{R}^{n+l}$. Let $F \in \mathcal{C}^1(U, \mathbb{R}^l)$. Let $c \in \mathbb{R}^l$. Suppose that there exists $a = (a_1, \dots, a_{n+l}) \in \mathbb{R}^{n+l}$ is such that $F(a) = c$ and

$$\begin{vmatrix} \frac{\partial F_1}{\partial x_{n+1}} \Big|_a & \cdots & \frac{\partial F_1}{\partial x_{n+l}} \Big|_a \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_{n+1}} \Big|_a & \cdots & \frac{\partial F_m}{\partial x_{n+l}} \Big|_a \end{vmatrix} \neq 0$$

Then there exists a neighborhood $N \subseteq \mathbb{R}^n$ of $(a_1, \dots, a_n)$ and $g \in \mathcal{C}^1(N, \mathbb{R}^l)$ such that the following are true.

- $g(a_1, \dots, a_n) = (a_{n+1}, \dots, a_{n+l})$
- $\{(x, g(x)) \mid x \in N\} \subset U$
- $F(x, g(x)) = c$ for all $x \in N$.
- The Jacobian of g is given by

$$J_g(x) = - \begin{pmatrix} \frac{\partial F_1}{\partial x_{n+1}} \Big|_x & \cdots & \frac{\partial F_1}{\partial x_{n+l}} \Big|_x \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_{n+1}} \Big|_x & \cdots & \frac{\partial F_m}{\partial x_{n+l}} \Big|_x \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial F_1}{\partial x_1} \Big|_x & \cdots & \frac{\partial F_1}{\partial x_n} \Big|_x \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} \Big|_x & \cdots & \frac{\partial F_m}{\partial x_n} \Big|_x \end{pmatrix}$$

for all $x \in N$

Beware that more often we seen that the invertible matrix is not necessarily on the right hand side of ∂F . It could consist of multiple columns of ∂F simply because of the ordering of the variables which is completely by the writer's choice. Moreover, there may be more than one g to convert the implicit function into an explicit one if you consider different variables for the domain and the codomain.

3.3 Higher Order Derivatives

From here onwards, our function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ will be a scalar function else the Hessian Matrix cannot be defined. Do note that general second order differential operators for $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ do exists. It is just that we will not discuss it here.

Definition 3.3.1 (Second Order Partial Derivatives) Suppose that $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable with partial derivative operator $\partial_j f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ for $1 \leq j \leq n$. Define the second order partial derivative at $x_0 \in U$ to be

$$\partial_{ij} f(x_0) = \frac{\partial}{\partial x_i} \partial_j f(x) \Big|_{x=x_0} = \frac{\partial^2}{\partial x_i \partial x_j} f(x) \Big|_{x=x_0}$$

for $1 \leq i \leq n$ if it exists. This is done by treating $\partial_j f$ as a function from $\mathbb{R}^n$ to $\mathbb{R}$ and taking the partial derivative of it.

Definition 3.3.2 (Hessian Matrix) Suppose that all second order partial derivatives of $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ exists. Define the Hessian Matrix f to be

$$H_f(x) = \partial^2 f(x) = \begin{pmatrix} \partial_{11}f(x) & \cdots & \partial_{1n}f(x) \\ \vdots & \ddots & \vdots \\ \partial_{n1}f(x) & \cdots & \partial_{nn}f(x) \end{pmatrix}$$

With $f : \mathbb{R}^n \rightarrow \mathbb{R}$, to say that Df is differentiable means that there exists a linear operator T such that

$$\lim_{h \rightarrow 0} \frac{|Df(x+h) - Df(x) - T(h)|}{|h|} = 0$$

as defined by the definition of differentiability. Notice that $Df(x) \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$. This means that T must take the form $T : \mathbb{R}^n \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$ so that $T(h) \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$ makes sense. Thus $T \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}))$. This leads to the following identification:

Proposition 3.3.3 Let $n \in \mathbb{N} \setminus \{0\}$. Then

$$\text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})) \cong \text{Hom}_{\mathbb{R}}^2(\mathbb{R}^n, \mathbb{R})$$

where $\text{Hom}_{\mathbb{R}}^2(\mathbb{R}^n, \mathbb{R})$ is the space of all bilinear forms $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$.

Proof Let $T \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}))$. Then for every $x \in \mathbb{R}^n$, $T(x) : \mathbb{R}^n \rightarrow \mathbb{R}$. Thus $T(x)(y) \in \mathbb{R}$ with $x, y \in \mathbb{R}^n$. Obviously we can write it into $T(x)(y) = B(x, y)$. Conversely any bilinear form can be written into the above form. In particular they both are vector spaces of dimension n^2 . From linear algebra we know that $B(x, y) = x^T A y$ for some matrix A . If $T(x) = Cx$ with C representing T , then

$$B(x, y) = (T(x)) \cdot y = (Cx)^T y = x^T C^T y$$

■

Proposition 3.3.4 Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable for all $x \in U$ and $Df : U \subseteq \mathbb{R}^n \rightarrow \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$ be differentiable. Then $D(Df)(x) = H_f(x)$. Moreover, $H_f(x)$ is symmetric and all second order partial derivatives commute.

Lemma 3.3.5 If all second order partial derivatives of $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ at x is continuous for $x \in U$, then

$$\frac{\partial^2}{\partial x_i \partial x_j} f(x) = \frac{\partial^2}{\partial x_j \partial x_i} f(x)$$

Definition 3.3.6 (Twice differentiable and Continuous derivatives) Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ where

$$f(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_m(x) \end{pmatrix} \text{ with } f_k(x) : \mathbb{R}^n \rightarrow \mathbb{R} \text{ for } k \in \{1, \dots, m\}. \text{ Define}$$

$$\mathcal{C}^2(U, \mathbb{R}^m) = \{f : U \rightarrow \mathbb{R}^m \mid \partial^2 f_k(x) : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R} \text{ is continuous for } k \in \{1, \dots, m\}\}$$

Theorem 3.3.7 [Second Order Taylor Expansion] Let $U \subset \mathbb{R}^n$ be convex and $x, x+h \in U$. If $f \in \mathcal{C}^2(U)$ then

$$f(x+h) = f(x) + \sum_{k=1}^n h_k \frac{\partial f(x)}{\partial x_k} + \frac{1}{2} \sum_{i,j=1}^n h_i h_j \frac{\partial^2 f(x)}{\partial x_i \partial x_j} + R(h)$$

where

$$\lim_{h \rightarrow 0} \frac{|R(h)|}{|h|^2} = 0$$

3.4 Second Order Derivative Test

Definition 3.4.1 (Critical Point) We say that $p \in U$ is a critical point of $f \in \mathcal{C}^1(U)$ if $\nabla f(p) = 0$.

Proposition 3.4.2 If f has a local minimum or maximum at p then $\nabla f(p) = 0$.

Theorem 3.4.3 (Second Order Derivative Test) Suppose that $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ such that $H_f(x)$ exists and is continuous ($f \in \mathcal{C}^2(U)$). Suppose that $\nabla f(p) = 0$ for some $p \in U$ (All first order derivatives are 0 at p or p is a critical point).

- If $x^T H_f(p) x > 0$ for all $x \in \mathbb{R}^n \setminus \{0\}$ then f has a strict local minimum at p
- If $x^T H_f(p) x < 0$ for all $x \in \mathbb{R}^n \setminus \{0\}$ then f has a strict local maximum at p
- If there exists $x, y \in \mathbb{R}^n \setminus \{0\}$ such that $x^T H_f(p) x > 0$ and $y^T H_f(p) y < 0$ then p is a saddle point
- The test is inconclusive otherwise.

4 Vector Calculus

4.1 Regions in Euclidean Space

Before we define integration of functions, we need to make sense of the domain of our integral.

Definition 4.1.1 (Regions) A region in $\mathbb{R}^n$ is a bounded open subset Ω of $\mathbb{R}^n$ such that there exists a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with the property that

- all partial derivatives of f are continuous
- $\Omega = \{x \in \mathbb{R}^n \mid f(x) < 0\}$
- If $p \in \mathbb{R}^n$ is such that $f(p) = 0$, then the partial derivatives of f do not simultaneously vanish.

Also define the boundary of a region, $\partial\Omega$ to be $\{x \in \mathbb{R}^n \mid f(x) = 0\}$.

Example 4.1.2

Let $R > 0$. Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $f(x, y, z) = x^2 + y^2 + z^2 - R^2$. Then $\{x \in \mathbb{R}^3 \mid f(x) < 0\}$ is a region in $\mathbb{R}^3$.

Proposition 4.1.3 Let $\Omega \subset \mathbb{R}^n$ be a region with boundary $f(x) = 0$. Then there exists $0 < l < n$ and $r : \mathbb{R}^l \rightarrow \mathbb{R}^n$ such that $f(r(x)) = 0$.

Proof Since $\nabla f(p) \neq 0$ for all $p \in \mathbb{R}^n$, we can apply the implicit function theorem to find function $g : \mathbb{R}^l \rightarrow \mathbb{R}^{n-l}$ such that $f(x, g(x)) = 0$ for any $p \in f^{-1}(0)$. Now define $r : \mathbb{R}^l \rightarrow \mathbb{R}^n$ by $r(x) = (x, g(x))$. Then clearly $f(r(x)) = 0$. ■

Example 4.1.4

Let $R > 0$. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function defined by $f(x, y, z) = x^2 + y^2 + z^2 - R^2$. Consider the region $\{x \in \mathbb{R}^3 \mid f(x) < 0\}$.

In particular, if $l = 1$, then we call r a curve. If $l = 2$, then r would be a surface. This is precisely the parametrization of a boundary, be it a curve or a surface.

4.2 Differentiation and Integration of Curves

This section is dedicated to vector functions, which are functions with domain $\mathbb{R}$ and codomain $\mathbb{R}^n$.

Definition 4.2.1 (Curves in Euclidean Space) Let $C \subseteq \mathbb{R}^n$ be a subset. We say that C is a curve if there exists a continuous function $r : I \rightarrow \mathbb{R}^n$ such that

$$C = \text{im}(r)$$

In this case we call r a parametrization of C .

Definition 4.2.2 (Simple and Closed Curves) Let $C \subseteq \mathbb{R}^n$ be a curve.

- C is simple if there exists a parametrization $r : I \rightarrow \mathbb{R}^n$ of C such that r is injective.
- C is closed if there exists a region $\Omega \subseteq \mathbb{R}^n$ such that $C = \partial\Omega$.

Definition 4.2.3 (Differentiable and Regular Parameterizations) Let $C \subseteq \mathbb{R}^n$ be a curve. Let $r : I \rightarrow \mathbb{R}^n$ be a parametrization of C .

- We say that $r \in \mathcal{C}^n$ if $r^{(k)}$ is differentiable for all $0 \leq k \leq n-1$, and $r^{(n)}$ is continuous.
- We say that r is regular if $|r'(t)| \neq 0$ for all $t \in I$.

Proposition 4.2.4 Let $r : I \rightarrow \mathbb{R}^n$ be a function. If $\text{im}(r)$ is simple, closed and r is regular, then there exists a region $\Omega \subseteq \mathbb{R}^n$ such that

$$\partial\Omega = \text{im}(r)$$

Definition 4.2.5 (Arc Length Function and Parametrization) Let $r : I \rightarrow \mathbb{R}^n$ be a function. Define the Arc Length Function by

$$s(t) = \int_{t_0}^t |\mathbf{r}'(u)| du$$

Define the Arc Length Parametrization by $\mathbf{r}(s)$.

Lemma 4.2.6 Let $r : I \rightarrow \mathbb{R}^n$ be a function. Denote the arc length function by $s(t)$. Then we have

$$|\mathbf{r}'(s(t))| = 1$$

Proof We show that $|\mathbf{r}'(s)| = 1$. By the chain rule,

$$\begin{aligned} \left| \frac{d}{ds} \mathbf{r}(s(t)) \right| &= \left| \frac{d\mathbf{r}}{dt} \cdot \frac{dt}{ds} \right| \\ &= \frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t)|} && \text{(By the FTC)} \\ &= 1 \end{aligned}$$

■

Definition 4.2.7 (Line Integrals on Scalar Functions) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Let $r : [a, b] \rightarrow \mathbb{R}^n$ a smooth curve with image C . Define the line integral along a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ to be

$$\int_C f dr = \int_a^b f(r(t)) |\mathbf{r}'(t)| dt$$

If r is a closed curve then we write the integral as $\oint_C f$.

dr in the integral is justified by noting that if r is a function of t , then $dr = \mathbf{r}'(t)dt$. But since dr is a positive line segment, we take the absolute value, giving us

$$dr = |dr| = |\mathbf{r}'(t)|dt = dt$$

There is no proper infinite sum that we can justify for $\int_C f$ because it is simply a notation short hand for the integral on the right hand side. Notice that the parameterization r does not appear on this notation because there can be multiple parametrizations with the same image.

4.3 Vector Fields

Definition 4.3.1 (Vector Fields) Let $U \subseteq \mathbb{R}^n$ be a subset. A vector field on U is a function $v : U \rightarrow \mathbb{R}^n$.

Definition 4.3.2 (Conservative Vector Fields) We say that a vector field $v : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ is conservative if there exists some scalar function $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$v = \nabla \phi$$

Definition 4.3.3 (Divergence) Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a vector field. Define the divergence of f to be

$$\nabla \cdot f = \left(\frac{\partial}{\partial x_1} \quad \cdots \quad \frac{\partial}{\partial x_n} \right) \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} = \sum_{k=1}^n \frac{\partial f_k}{\partial x_k}$$

Definition 4.3.4 (Line Integrals on Vector Fields) Let $v : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a vector field. Let $r : [a, b] \rightarrow \mathbb{R}^n$ be a curve with image $C \in U$. Define the integral of v along the curve C to be

$$\int_C v \cdot dr = \int_a^b v(r(t)) \cdot r'(t) dt$$

If r is a closed curve then we write the integral as $\oint_C v \cdot dr$.

Proposition 4.3.5 Let $v : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a vector field. Let $r : [a, b] \rightarrow \mathbb{R}^n$ be a curve with image $C \in U$. Then

$$\int_C v \cdot dr = \sum_{k=1}^n \int_C v_k dx_k$$

Proposition 4.3.6 If $v : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ is conservative with $v = \nabla f$ and $r : [a, b] \rightarrow \mathbb{R}^n$ is a curve with image C then

$$\int_C v \cdot dr = f(r(b)) - f(r(a))$$

Theorem 4.3.7 (Equivalent Characterization of Conservative Vector Fields) Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a path. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a vector field. Then the following are equivalent.

- F is conservative
- $\int_\gamma F(r) \cdot dr$ is independent of the choice of γ
- $\int_\gamma F(r) \cdot dr = 0$ if γ is closed

5 2 Dimensional Calculus

5.1 Double Integrals

Definition 5.1.1 (Positively Oriented Tangents and Outward Normals) Let Ω be a region in $\mathbb{R}^2$ with boundary $f(x) = 0$ and parametrization $r : \mathbb{R} \rightarrow \mathbb{R}^2$. We say that r is positively oriented if as the domain of r increases, the tangent vector $r'(t)$ moves anticlockwise. In this case, we say that $\begin{pmatrix} -r_2(p) \\ r_1(p) \end{pmatrix}$ for $p \in \partial\Omega$ is the outward normal vector which is perpendicular to the tangent space $T_p(\partial\Omega)$.

Definition 5.1.2 (Double Integrals) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Define the double integral over f on a region Ω to be

$$\iint_{\Omega} f(x, y) \, dx \, dy$$

Proposition 5.1.3 Let $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be integrable on a region Ω and $a, b \in \mathbb{R}$. Then the following are true for double integrals.

- $\iint_{\Omega} (af(x, y) + bg(x, y)) \, dA = a \iint_{\Omega} f(x, y) \, dA + b \iint_{\Omega} g(x, y) \, dA$
- If $f(x, y) \geq g(x, y)$ for all $x, y \in \Omega$, then $\iint_{\Omega} f(x, y) \, dA \geq \iint_{\Omega} g(x, y) \, dA$
- If $\Omega = D_1 \cup D_2$ and $D_1 \cap D_2 = \emptyset$, then $\iint_{\Omega} f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \iint_{D_2} f(x, y) \, dA$

Proposition 5.1.4 (Fubini's Theorem) If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous and bounded on the rectangle $R = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$. Then

$$\int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy$$

Proposition 5.1.5 If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function such that $f(x, y) = g(x)h(y)$ where $g, h : \mathbb{R} \rightarrow \mathbb{R}$ and $R = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$, then

$$\iint_R f(x, y) \, dA = \int_a^b g(x) \, dx \int_c^d h(y) \, dy$$

Theorem 5.1.6 Let R be a region. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $f(x, y) = \begin{pmatrix} u(x, y) \\ v(x, y) \end{pmatrix}$ such that $f|_R$ is a bijection and $\partial f \in \mathcal{C}^1(R)$. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then

$$\iint_S g(u, v) \, du \, dv = \iint_R g(x, y) |\partial f| \, dx \, dy$$

The conditions for f being a local bijection is given by the inverse function theorem.

Theorem 5.1.7 (Moments) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Let $\rho(x, y)$ give the moment at (x, y) . The moment about the x -axis on a region $D \subset \mathbb{R}^2$ is given by

$$M_x = \iint_D y \rho(x, y) \, dA$$

and the moment about the y -axis is given by

$$M_y = \iint_D x \rho(x, y) \, dA$$

Theorem 5.1.8 (Mass) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Let $\rho(x, y)$ give the moment at (x, y) . Let $D \subset \mathbb{R}^2$ be a region. Then the mass is given by

$$m = \iint_D \rho(x, y) dA$$

Theorem 5.1.9 (Center of Mass) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Let $\rho(x, y)$ give the moment at (x, y) . Let $D \subset \mathbb{R}^2$ be a region. The center of mass is given by

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right)$$

5.2 Green's Theorem for a Planar Region

Definition 5.2.1 (Curl) Let $v : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a continuously differentiable planar vector field given by $v(x, y) = (a(x, y), b(x, y))$. Define the curl of v to be

$$\text{curl}(v) = \frac{\partial b}{\partial x} - \frac{\partial a}{\partial y}$$

Theorem 5.2.2 (Green's Theorem for a Rectangular Region) Let $v : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a continuously differentiable planar vector field where $v(x, y) = \begin{pmatrix} v_1(x, y) \\ v_2(x, y) \end{pmatrix}$. Let $\Omega = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$ such that $\partial\Omega \subset U$. Then

$$\iint_{\Omega} \text{curl}(v) dA = \int_{\partial\Omega} v \cdot dr$$

where r is positively oriented.

Proof Clearly, we have

$$\iint_{\Omega} \text{curl}(v) dA = \int_a^b v_1(x, c) dx + \int_b^c v_2(d, y) dy + \int_c^a v_1(x, d) dx + \int_d^b v_2(a, y) dy$$

■

Theorem 5.2.3 (Green's Theorem for a Planar Region) Let $v : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a continuously differentiable planar vector field. Let Ω be a region such that $\Omega \cup \partial\Omega \subset U$ and r a parametrization of $\partial\Omega$. Then

$$\iint_{\Omega} \text{curl}(v) dA = \oint_{\partial\Omega} v \cdot dr$$

where r is positively oriented.

Theorem 5.2.4 (Conservative Implies Zero Curl) If $v : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a conservative vector field with continuous partial derivatives on U , then

$$\text{curl}(v) = 0$$

5.3 Divergence Theorem for a Planar Region

Definition 5.3.1 (Flux across a Curve) Let $v : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a planar vector field. Let $\Omega \subset \mathbb{R}^2$ be a region with $\partial\Omega$ parametrized by r . Define the flux of v across Ω to be

$$\text{flux of } v = \int_{\partial\Omega} v \cdot n dr$$

where r is positively oriented such that n is the outward unit normal.

Theorem 5.3.2 (Divergence Theorem for a Planar Region) Let $v : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a continuously differentiable planar vector field. Let Ω be a region such that $\Omega \cup \partial\Omega \subset U$ and r a parametrization of $\partial\Omega$. Then

$$\iint_{\Omega} \nabla \cdot v \, dA = \int_{\partial\Omega} v \cdot n \, dr$$

where r is positively oriented such that n is the outward unit normal.

6 3 Dimensional Calculus

6.1 Triple Integrals

Definition 6.1.1 (Triple Integral) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a function. Define the triple integral of f in a region Ω to be

$$\iiint_{\Omega} f(x, y, z) \, dx \, dy \, dz$$

Theorem 6.1.2 Let $R \subset \mathbb{R}^3$ be a region. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $f(x, y, z) = \begin{pmatrix} u(x, y, z) \\ v(x, y, z) \\ w(x, y, z) \end{pmatrix}$ such that $f|_R$ is a bijection and $\partial f \in \mathcal{C}^1(R)$. Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}$. Then

$$\iiint_S g(u, v, w) \, du \, dv \, dw = \iiint_R g(x, y, z) |\partial f| \, dx \, dy \, dz$$

Theorem 6.1.3 (Cylindrical Coordinates) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$. Let $E \subset \mathbb{R}^3$ be a region. Then

$$\iiint_R f(x, y, z) \, dV = \iiint_R f(r \cos \theta, r \sin \theta, z) r \, dr \, d\theta \, dz$$

with $x = r \cos(\theta)$ and $y = r \sin(\theta)$ and $z = z$

Theorem 6.1.4 (Spherical Coordinates) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$. Let $E \subset \mathbb{R}^3$ be a region. Then

$$\iiint_R f(x, y, z) \, dV = \iiint_R f(r \cos \theta \sin \phi, r \sin \theta \sin \phi, r \cos \phi) r^2 \sin(\phi) \, dr \, d\theta \, d\phi$$

with $x = r \sin(\phi) \cos(\theta)$ and $y = r \sin(\phi) \sin(\theta)$ and $z = r \cos(\phi)$

Theorem 6.1.5 (Mass) Consider a solid occupying a region $\Omega \subset \mathbb{R}^3$ with density function $\rho(x, y, z)$. Then its total mass is

$$M = \iiint_{\Omega} \rho(x, y, z) \, dV$$

Theorem 6.1.6 (Center of Mass) Consider a solid occupying a region $\Omega \subset \mathbb{R}^3$ with density function $\rho(x, y, z)$ and total mass M . Then

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{1}{M} \iiint_{\Omega} \rho x \, dV, \frac{1}{M} \iiint_{\Omega} \rho y \, dV, \frac{1}{M} \iiint_{\Omega} \rho z \, dV \right)$$

6.2 Surface Integrals

Theorem 6.2.1 Let S be a surface parametrized by $\mathbf{r}(u, v)$. The surface area is given by

$$A = \iint_{\Omega} |\mathbf{r}_u \times \mathbf{r}_v| \, du \, dv$$

where Ω is the parameter domain.

Lemma 6.2.2 The formula of the area of a surface given by $z = f(x, y)$ where $(x, y) \in R \subset \mathbb{R}^2$ is

given by

$$A = \iint_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dx dy$$

6.3 Divergence Theorem

Definition 6.3.1 (Curl) Let $f : U \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field. Then the curl of f is defined as

$$\nabla \times f = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

Proposition 6.3.2 If $v : U \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a conservative vector field with continuous partial derivatives on U , then

$$\text{curl}(v) = 0$$

Definition 6.3.3 (Flux of a Surface) Let $v : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field. Let $\Omega \subset \mathbb{R}^3$ be a volume. Define the flux of v across Ω to be

$$\text{flux of } v = \iint_{\partial\Omega} v \cdot n dA$$

where n is the outward pointing normal.

Lemma 6.3.4 (Practical Definition of Flux) Let $\mathbf{v} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Let $\Omega \subset \mathbb{R}^3$. Let $r : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ parameterize the surface $\partial\Omega$. Then

$$\iint_{\partial\Omega} v \cdot n dA = \iint_{\partial\Omega} \mathbf{v}(r(u, v)) \cdot \left(\frac{\partial r}{\partial u} \times \frac{\partial r}{\partial v} \right) du dv$$

Theorem 6.3.5 (The Divergence Theorem) Let $v : U \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a C^1 vector field. Let Ω be a region in $\mathbb{R}^3$ with $\partial\Omega$ parametrized by r such that $\Omega \cup \partial\Omega \subset U$. Then

$$\iiint_{\Omega} \nabla \cdot v dV = \iint_{\partial\Omega} v \cdot n dA$$

where r is positively oriented such that n is the outwards unit normal.

6.4 Stoke's Theorem

Definition 6.4.1 (Pointwise Circulation) Let $\mathbf{F}(x, y, z)$ give the velocity of a fluid. The pointwise circulation of $\mathbf{F}$ is given by

$$\nabla \times \mathbf{F} \cdot \hat{n}$$

Theorem 6.4.2 (Net Circulation) The net circulation over a surface S is given by

$$\iint_S \nabla \times \mathbf{F} \cdot \hat{n} dS$$

Theorem 6.4.3 (Stoke's Theorem) Let $\mathbf{F}(x, y, z)$ be a vector field. Let S be a surface with unit normal $\hat{n}$ and boundary curve C , oriented according to the right hand rule. Then

$$\iint_S \nabla \times \mathbf{F} \cdot \hat{n} dS = \int_C \mathbf{F} \cdot d\mathbf{r}$$